

Haasith Venkata Sai Pasala

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EDUCATION

International Institute of Information Technology Hyderabad (IIIT-H) <i>M.S. by Research in Electronics and Communication Engineering (Robotics)</i>	Hyderabad, India Aug. 2021 – Jun. 2024
Anil Neerukonda Institute of Technology and Sciences <i>Bachelor of Engineering in Electronics and Communication Engineering</i>	Vishakapatnam, India Aug. 2014 – May 2018

TECHNICAL SKILLS

Languages: C/C++, Python, Java, JavaScript, C
Autonomy & Perception: ROS 1 / ROS 2 (Foxy, Humble), Nav2, SLAM (FAST-LIO, D-LIO, RTAB-Map), OpenCV, PointNet++, Sensor Fusion (GPS, Lidar), Visual Servoing, Trajectory Optimization
Simulation & Tools: Docker, AWS EC2, Gazebo, PyBullet, Git, CMake, Linux
Embedded & Hardware: Embedded Linux, ARM Cortex, ALSA, I2C, SPI, Unitree Go2W, XArm7
Testing & QA: Appium, Selenium, Cucumber (BDD)

EXPERIENCE

Technical Lead (Consultant) <i>Stealth Mode Company</i>	Jul. 2025 – Present <i>Remote</i>
- Engineered the complete autonomy stack and environment setup for a Unitree Go2W quadruped robot, migrating the control architecture between ROS 2 Foxy and Humble to ensure hardware driver compatibility. - Designed and integrated a robust state estimation pipeline utilizing FAST-LIO, D-LIO, and RTAB-Map for precise localization, fusing GPS data for global positioning in outdoor environments. - Configured Nav2 and implemented whole-body control frameworks to enable autonomous navigation, obstacle avoidance, and complex legged locomotion. - Containerized the full simulation and deployment stack using Docker and provisioned AWS EC2 instances for scalable, cloud-based testing prior to hardware deployment.	
Embedded Software Engineer <i>Miko</i>	Jun. 2024 – Jun. 2025 <i>Mumbai, India</i>
- Re-architected the "Miko Dance" feature for offline execution by replacing WebSocket-dependent network calls with localized C++ processing logic, optimizing system responsiveness. - Led hardware-software system bring-up and validation, systematically debugging critical latency, synchronization, and audio quality anomalies across the entire tech stack. - Developed and integrated ALSA-based audio subsystems on ARM embedded Linux, enabling highly reliable audio interaction for production-level consumer robots. - Implemented low-level C/C++ software drivers to configure the TAS2563 audio amplifier, managing communication via I2C and I2S/TDM interfaces.	
Research Assistant <i>Robotics Research Centre, IIIT-Hyderabad</i>	Sep. 2021 – Jun. 2024 <i>Hyderabad, India</i>
- Engineered an advanced perception pipeline for throwing manipulation by deploying PointNet++ for 3D point cloud analysis and integrating ensemble learning (SVM/Random Forests) for robust object recognition. - Formulated rigid body dynamics and implemented trajectory optimization algorithms using MATLAB, Gazebo, and ROS to compute optimal initial conditions for dynamic object throwing. - Designed and integrated a novel hybrid end-effector with an XArm7 robotic arm and mobile base, developing the control architecture using ROS, Raspberry Pi, and Arduino. - Conducted two-stage parameter identification and applied optimization techniques to bridge mathematical models with physical hardware, drastically minimizing sim-to-real discrepancies.	

Consultant Engineer

Mar. 2019 – Jun. 2020

L&T Technology Services

Mysore, India

- Engineered end-to-end automation testing frameworks for a regulated mobile medical application (Eli Lilly) across two major release cycles, utilizing BDD principles with Cucumber.
- Designed and executed cross-platform automated testing pipelines for iOS and Android environments (real devices and emulators) utilizing Appium, WebdriverIO, and Selenium.
- Programmed scalable test features and step definitions in Java (TestNG) and JavaScript, ensuring high reliability and strict compliance for a critical healthcare application.

PROJECTS

Stereo SLAM System Development | *C++, OpenCV, Pose-Graph Optimization* Sep. 2021 – Dec. 2021

- Implemented Stereo Reconstruction, Perspective-n-Point (PnP), and Bundle Adjustment to develop a robust 1D and 2D SLAM pipeline.
- Applied Pose-Graph Optimization algorithms to correct trajectory drift and significantly enhance overall localization accuracy.

Sim-to-Real Validation of an Anthropomorphic Hand | *MuJoCo, 3D Printing, PID Control* [Dates]

- Developed a high-fidelity physics model in MuJoCo to simulate contact dynamics and underactuated grasping synergies for an anthropomorphic robotic hand.
- Led the physical realization phase, utilizing 3D printing for rapid prototyping and hardware assembly of the tendon-driven actuation system.
- Designed and executed experimental validation protocols, utilizing PID control to analyze the correlation between simulated grasp stability and real-world performance.

Eye-in-Hand Visual Servoing for Dynamic Tracking | *OpenCV, ROS, C++/Python* Jun. 2023

- Developed an active visual servoing control loop using an XArm manipulator equipped with an end-effector-mounted RGB-D camera.
- Utilized OpenCV and ROS TF to estimate the 6D pose of ArUco markers, computing the necessary velocity commands to dynamically adjust the robot's trajectory in real-time.

GELLO Framework for XArm Teleoperation | *Robotics, Kinematics, ROS* Nov. 2023

- Led the integration and deployment of the GELLO teleoperation framework for an XArm manipulator, architecting the end-to-end system for robust data collection and imitation learning.
- Configured a low-latency leader-follower control system, ensuring precise joint-space mapping and real-time responsiveness for complex remote manipulation tasks.

PUBLICATIONS & RECOGNITION

Identification and Learning-Based Control of an End-Effector for Targeted Throwing | *IEEE RA-L* Aug. 2024

- Accepted to the IEEE Robotics and Automation Letters (RA-L).
- Featured on **IEEE Spectrum** (Oct. 2024), highlighting the multimedia extension and real-world throwing manipulation capabilities.

A Novel Hybrid Gripper Capable of Grasping and Throwing Manipulation | *IEEE/ASME T-MECH* 2023

- Published in the IEEE/ASME Transactions on Mechatronics (Impact Factor: 6.4).
- Featured on **IEEE Spectrum** (Apr. 2023) for pioneering a dual-action mechanism bridging grasping and dynamic throwing.